

Set Theory for Computer Science

Week 3: Ordering Relations

Partial and linear ordering relations

Hasse diagrams

Cartesian and lexicographic order

Minimal and maximal elements

Partial ordering relations

A **partial order** on V is a relation of type $V \times V$ that is reflexive and transitive and anti-symmetric

Reminder

- ▶ reflexivity $x R x$
- ▶ transitivity $x R y \wedge y R z \rightarrow x R z$
- ▶ anti-symmetry $(x \neq y \wedge x R y) \rightarrow \neg y R x$
 $x R y \wedge y R x \rightarrow x = y$



Examples: natural numbers and sets

Natural numbers

The relation $\leq$ is a partial order on $\mathbb{N}$:

1. $\forall n: \quad n \leq n$ (reflexivity) ✓
2. $\forall k, m, n: \quad k \leq m \wedge m \leq n \rightarrow k \leq n$ (transitivity) ✓
3. $\forall m, n: \quad m \leq n \wedge n \leq m \rightarrow m = n$ (anti-symmetry) ✓

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Power sets

The **power set** $\mathcal{P}(V)$ of a set V is the set of all subsets of V

Example

$$V := \{1, 2\} \quad \Rightarrow \quad \mathcal{P}(V) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$$

Examples: natural numbers and sets

Natural numbers

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Power sets

The relation $\subseteq$ is a partial order on $\mathcal{P}(V)$:

1. $\forall A: \quad A \subseteq A$ (reflexivity) ✓
2. $\forall A, B, C: \quad A \subseteq B \wedge B \subseteq C \rightarrow A \subseteq C$ (transitivity) ✓
3. $\forall A, B: \quad A \subseteq B \wedge B \subseteq A \rightarrow A = B$ (anti-symmetry) ✓

Linear ordering relations

Consider a partial order R on a set V . Then

- ▶ $x, y \in V$ are **comparable** if $x R y$ or $y R x$
- ▶ R is a **linear** (or **total**) **order** if *all* $x, y \in V$ are comparable

Linear ordering relations

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Examples

Relation $\leq$ on $\mathbb{N}$:

for all n and m in $\mathbb{N}$
 $n \leq m$ or $m \leq n$

hence **total**

Relation $\subseteq$ on $\mathcal{P}(\{1, 2\})$:

$\{1\}$ is not a subset of $\{2\}$
 $\{2\}$ is not a subset of $\{1\}$

hence **not total**

Strict partial ordering relations

Consider a partial order R on a set V . Then

- ▶ the **strict partial order** S corresponding to R is defined by

$$x S y \iff x R y \text{ and } x \neq y$$

- ▶ in set language:
$$S = R \setminus \{\langle x, x \rangle : x \in V\}$$
$$R = S \cup \{\langle x, x \rangle : x \in V\}$$

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Theorem

A strict partial order is **irreflexive** and transitive and anti-symmetric

$$\forall x \neg(x S x)$$

Proof

- ▶ all pairs $\langle x, x \rangle$ have been removed from R
- ▶ but they are irrelevant for transitivity + anti-symmetry **together**

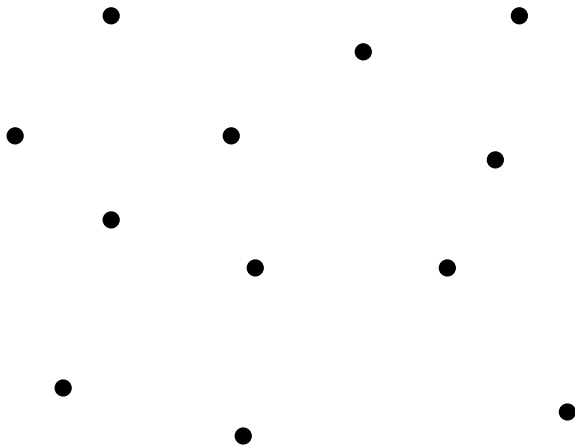
Partial and linear ordering relations

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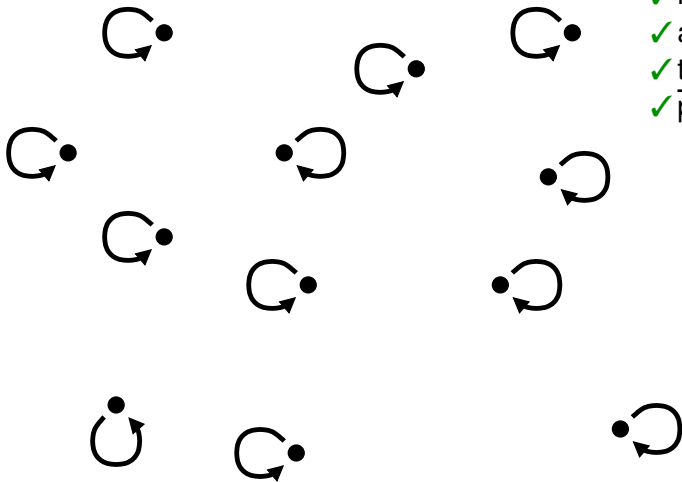
What does a partial order look like?



- ✗ reflexive
- ✓ anti-symm.
- ✓ transitive
- ✗ partial order

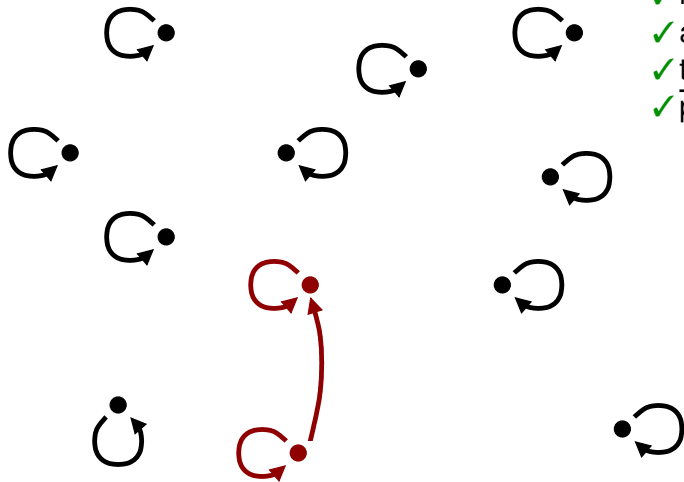
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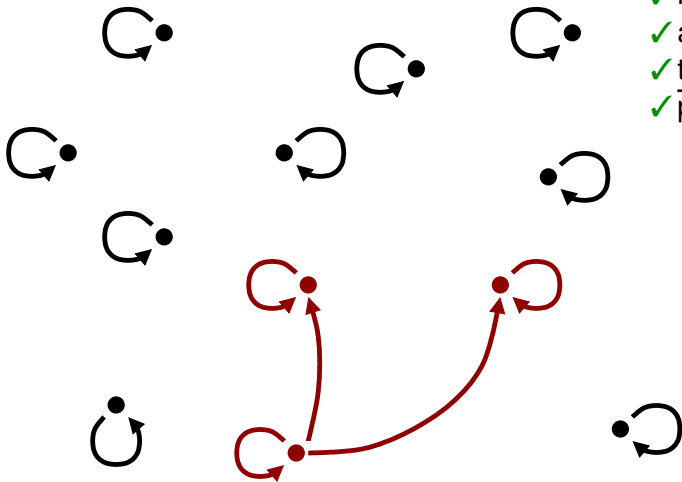
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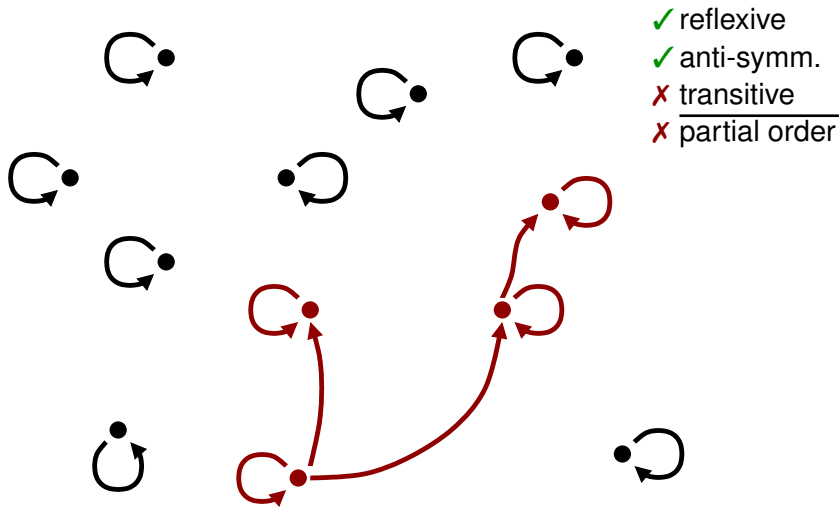


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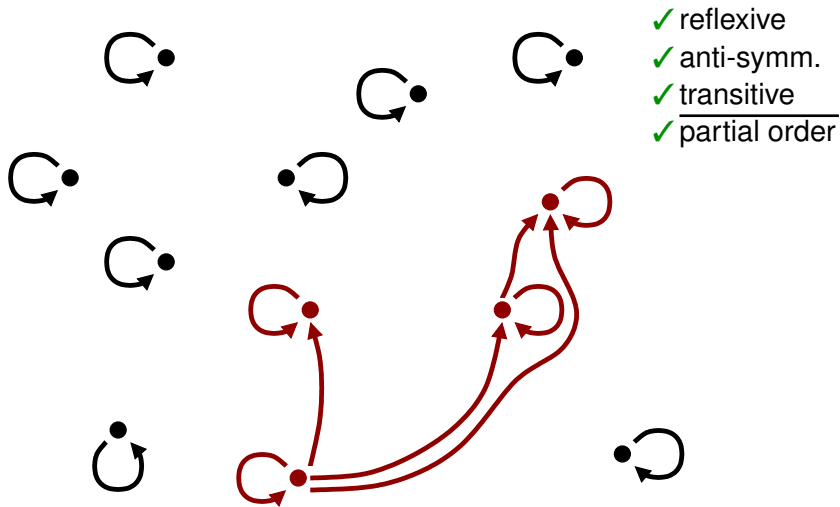
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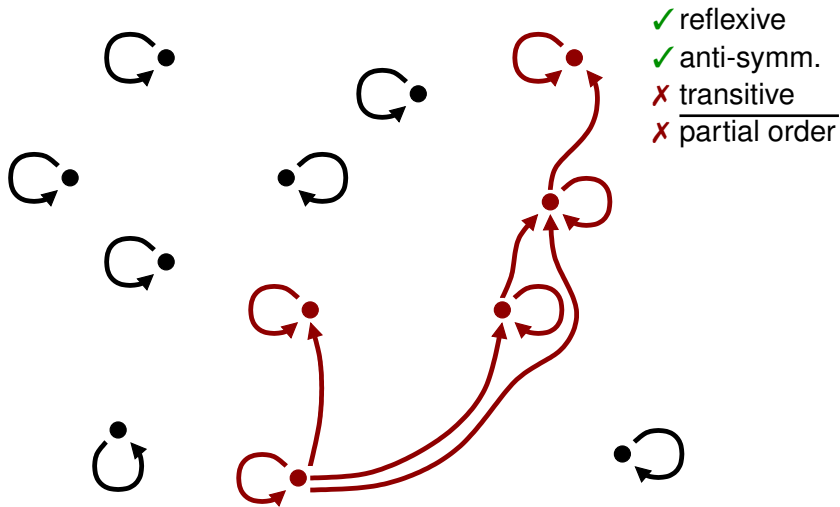
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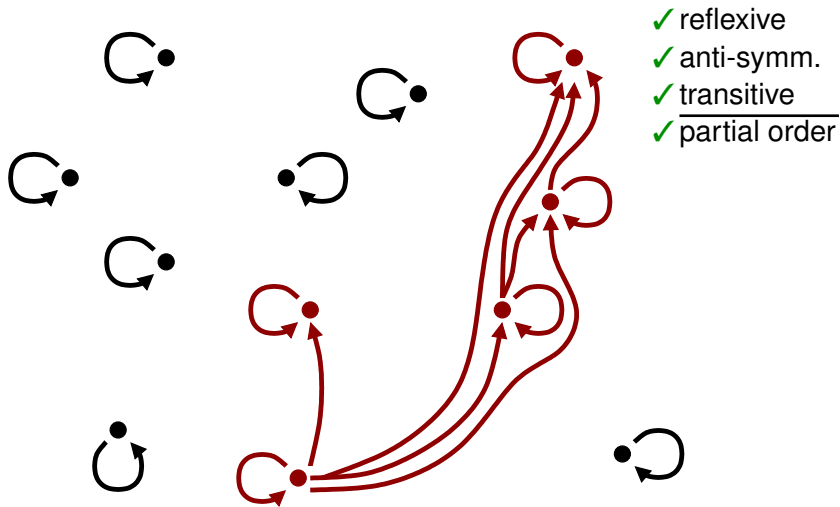
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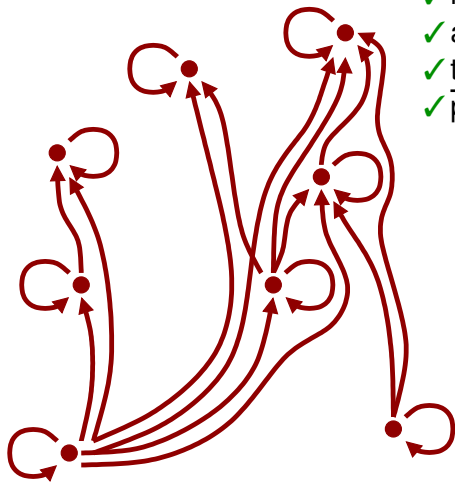
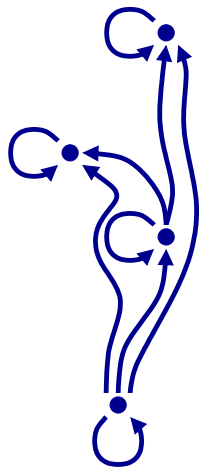
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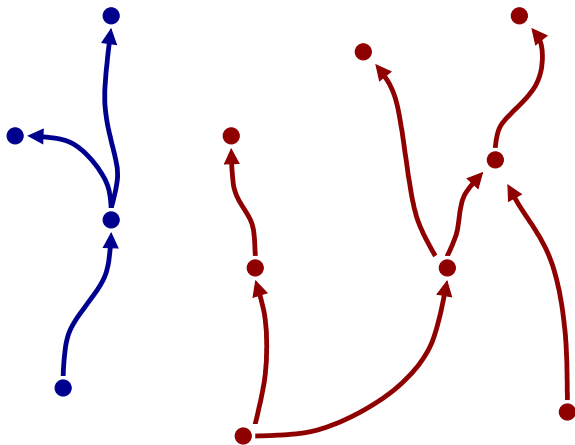
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What does a partial order look like?

Hasse diagram: omit arrows implied by reflexivity and transitivity



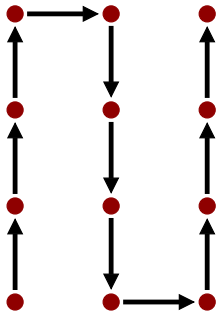
Elements form **chains** that can split & merge

What does a partial order look like?

Hasse diagram: omit arrows implied by reflexivity and transitivity

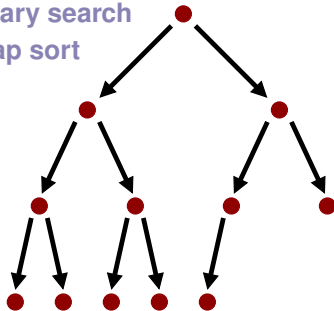
Two special examples

Linear order



Binary (min) heap

- ▶ binary search
- ▶ heap sort



How to construct a Hasse diagram?

Algorithm (Hasse diagram for a given partial order R on V)

1. for all $x \in V$ $G_x := \{y : y \neq x \text{ and } x R y\}$
2. for all $x \in V$ $H_x := G_x \setminus \bigcup_{y \in G_x} G_y$
3. for all $x \in V$ draw an arrow from x to every $y \in H_x$

($\bigcup_{y \in G_x} G_y$ is the **union** of all sets G_y with $y \in G_x$)

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Example $V := \{1, 2, 3, 4, 6, 12\}$ $R := \text{IsDivisorOf}$

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$R := \text{IsDivisorOf}$

$$G_1 = \{2, 3, 4, 6, 12\}$$

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$$G_3 = \{6, 12\}$$

$$G_4 = \{12\}$$

$$G_6 = \{12\}$$

$$G_{12} = \{\}$$

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$$H_1$$

$$G_2 = \{4, 6, 12\}$$

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$$H_1 = \{2, 3\}$$

$$G_2 = \{4, 6, 12\}$$

$$G_3 = \{6, 12\}$$

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remove $\bigcup_{y \in G_1} G_y$ from G_1

$$G_6 = \{12\}$$

$$G_{12} = \{\}$$

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remove $\bigcup_{y \in G_2} G_y$ from G_2

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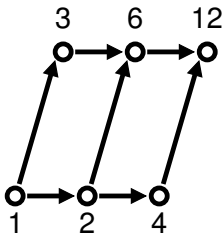
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Exercise

Construct and draw a Hasse diagram for the partial order $\subseteq$ on the set $\mathcal{P}(\{a, b, c\}) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}$.

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$$G_{\emptyset} = \{\{a\}, \{b\}, \{c\}, \{a, b\}, \{b, c\}, \{a, c\}, \{a, b, c\}\}$$

$$G_{\{a\}} = \{\{a, b\}, \{a, c\}, \{a, b, c\}\}$$

$$G_{\{b\}} = \{\{a, b\}, \{b, c\}, \{a, b, c\}\}$$

$$G_{\{c\}} = \{\{a, c\}, \{b, c\}, \{a, b, c\}\}$$

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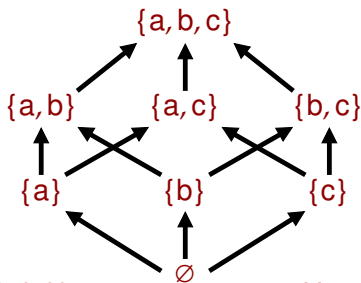
$$H_{\{c\}} = \{\{a, c\}, \{b, c\}\}$$

$$H_{\{a, b, c\}} = \{\}$$

Review exercises

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$$H_{\{c\}} = \{\{a, c\}, \{b, c\}\}$$

$$H_{\{a, b, c\}} = \{\}$$

break

Partial and linear ordering relations

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Cartesian and lexicographic order

Minimal and maximal elements

How to order a product set?

Suppose we have partial orders $\leq_A$ on A and $\leq_B$ on B

Problem: What is a natural order on the set $A \times B$?

How to order a product set?

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Problem: What is a natural order on the set $A \times B$?

Answer 1: Lexicographic order (A takes priority over B)

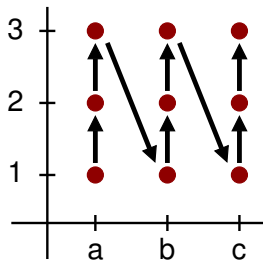
$$\langle a_1, b_1 \rangle \leq \langle a_2, b_2 \rangle \iff (a_1 <_A a_2) \vee (a_1 = a_2 \wedge b_1 \leq_B b_2)$$

Example

$$A = \{a, b, c\}$$

$$B = \{1, 2, 3\}$$

with the usual order



How to order a product set?

Suppose we have partial orders $\leq_A$ on A and $\leq_B$ on B

Problem: What is a natural order on the set $A \times B$?

Answer 2: Cartesian order (A and B treated equally)

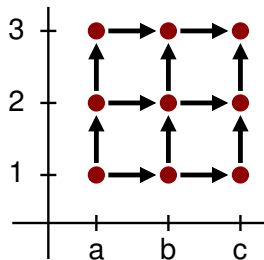
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NB Cartesian order is **not** total!

Cartesian order is a partial order

Theorem

The Cartesian order on $A \times B$ is a partial order

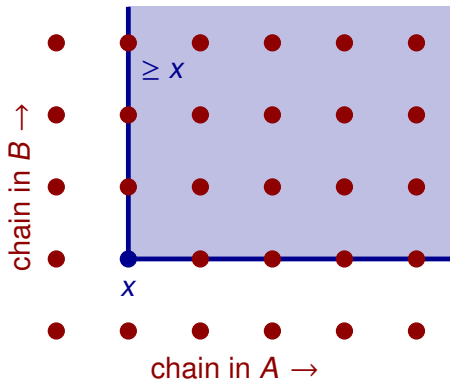
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Proof (in pictures)

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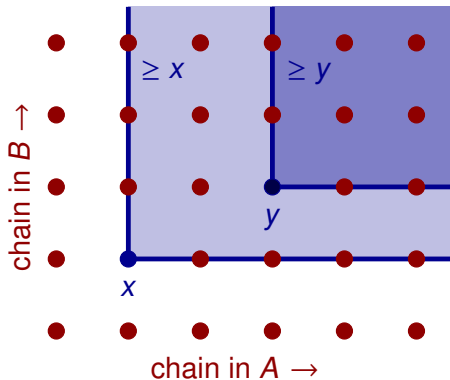
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Lexicographic order is a (total) partial order

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The lexicographic order on $A \times B$ is a partial order

Moreover, this order is total if $\leq_A$ and $\leq_B$ are both total

Lexicographic order is a (total) partial order

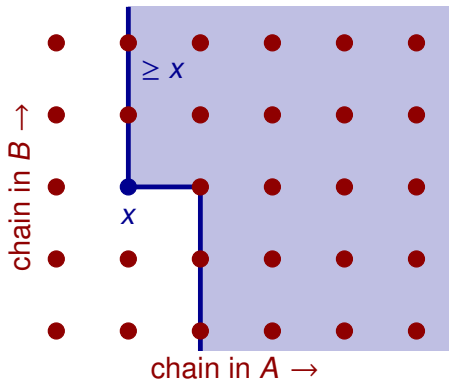
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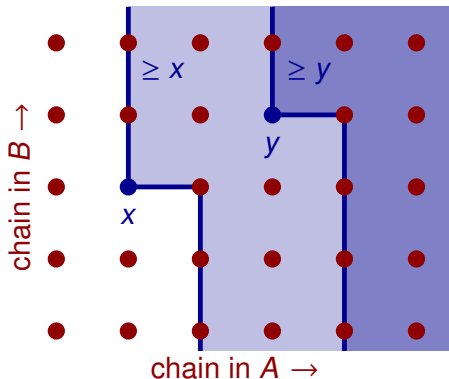
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4. totality: $x \leq y \vee y \leq x$

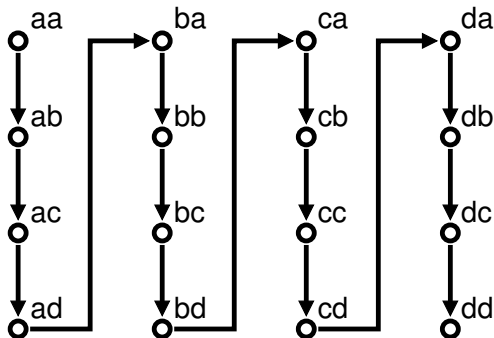


Lexicographic order = dictionary order

Example

Take $A := \{a, b, c, d\}$ with $\leq_A$ the usual alphabetic ordering of letters

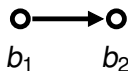
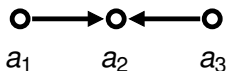
Lexicographic order on $A \times A$:



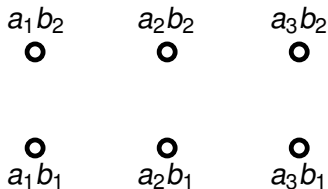
Review exercises

Exercise

Let $A := \{a_1, a_2, a_3\}$ and $B := \{b_1, b_2\}$ with partial orders $\leq_A$ and $\leq_B$ on A and B described by the following Hasse diagrams:



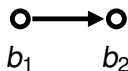
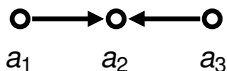
Draw Hasse diagrams for the Cartesian and lexicographic orders on $A \times B$ (induced by $\leq_A$ and $\leq_B$) by adding arrows to this picture:



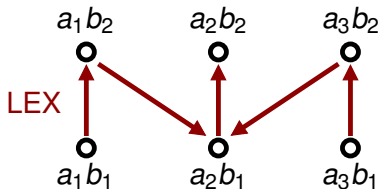
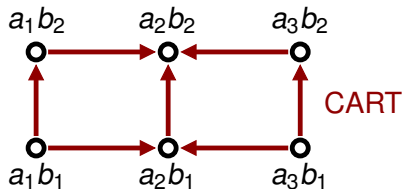
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Draw Hasse diagrams for the Cartesian and lexicographic orders on $A \times B$ (induced by $\leq_A$ and $\leq_B$) by adding arrows to this picture:



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Hasse diagrams

Cartesian and lexicographic order

Minimal and maximal elements

Minimal and maximal elements

Let $(V, \leq)$ be a partially ordered set, $A \subseteq V$, $m \in A$. Then m is a

- ▶ **largest element** of A if $\forall a \in A: a \leq m$
- ▶ **smallest element** of A if $\forall a \in A: m \leq a$
- ▶ **maximal element** of A if $\forall a \in A: (m \leq a \rightarrow m = a)$
- ▶ **minimal element** of A if $\forall a \in A: (a \leq m \rightarrow a = m)$

Terminology:

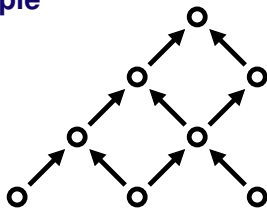
- ▶ largest element = **maximum**
- ▶ smallest element = **minimum**

Minimal and maximal elements

Let $(V, \leq)$ be a partially ordered set, $A \subseteq V$, $m \in A$. Then m is a

- ▶ **largest element** of A if $\forall a \in A: a \leq m$
- ▶ **smallest element** of A if $\forall a \in A: m \leq a$
- ▶ **maximal element** of A if $\forall a \in A: (m \leq a \rightarrow m = a)$
- ▶ **minimal element** of A if $\forall a \in A: (a \leq m \rightarrow a = m)$

Example

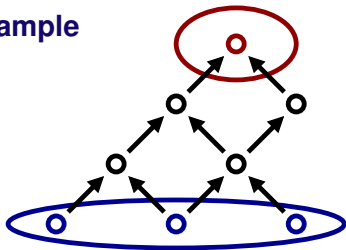


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Example



largest element (maximum)

minimal elements

Theorem 1: Every maximum of A is a maximal element of A

Theorem 2: If A has a maximum, it is the *only* maximal element

Properties of maximal elements: proofs

Theorem 1: Every maximum of A is a maximal element of A

Theorem 2: If A has a maximum, it is the *only* maximal element

Proof of Theorem 1

Suppose m is a maximum of A . Let $a \in A$ be arbitrary. Then

$$\forall x \in A: x \leq m \quad \Rightarrow \quad a \leq m$$

so if $m \leq a$, then $m = a$ (anti-symmetry of $\leq$). Therefore

$$\forall a \in A: (m \leq a \rightarrow m = a)$$

Hence m is a maximal element

Properties of maximal elements: proofs

Theorem 1: Every maximum of A is a maximal element of A

Theorem 2: If A has a maximum, it is the *only* maximal element

Proof of Theorem 2

Suppose m is a maximum and p is a maximal element of A . Then

$$\begin{array}{ll} \forall a \in A: a \leq m & \Rightarrow \quad p \leq m \\ \forall a \in A: (p \leq a \rightarrow p = a) & \Rightarrow \quad p \leq m \rightarrow p = m \end{array}$$

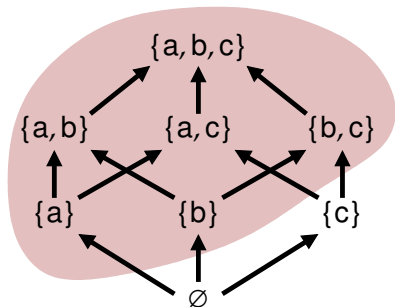
Therefore $p = m$

Review exercises

Exercise

Consider the Hasse diagram for $\subseteq$ on the set $\mathcal{P}(\{a, b, c\})$.

- (a) Find and classify all extremal elements of $\mathcal{P}(\{a, b, c\})$.



Let $A := \mathcal{P}(\{a, b, c\}) \setminus \{\emptyset, \{c\}\}$ (the highlighted part in the picture).

- (b) Find and classify all extremal elements of A .

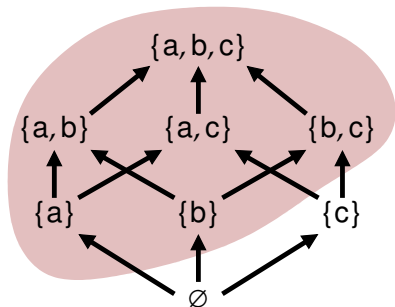
Review exercises

Exercise

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$\{a, b, c\}$ maximum
 $\emptyset$ minimum



Let $A := \mathcal{P}(\{a, b, c\}) \setminus \{\emptyset, \{c\}\}$ (the highlighted part in the picture).

- (b) Find and classify all extremal elements of A .

$\{a, b, c\}$ maximum
 $\{a\} \{b\}$ minimal elements